SEQUENCE 11

Cross-tabulating two qualitative variables

1–12 March 2027

11.1 Two-way tables and marginal frequencies

Key point · Definitions. A qualitative variable describes a category. It is nominal if its categories have no natural order (means of transport), ordinal if they are ordered (level of satisfaction). A two-way table counts individuals according to two variables. The sums of the rows and columns are the margins. A marginal frequency divides a margin by the total count.

Example 1. In a population of 100 individuals, two characteristics A, B are observed:

	A	$\overline{A}$	Total
B	20	15	35
$\overline{B}$	12	53	65
Total	32	68	100

Calculate the marginal frequencies of A, B, $\overline{B}$, then the frequency of individuals satisfying both A and B.

Solution. $f(A) = 32/100 = 0.32$; $f(B) = 35/100 = 0.35$; $f(\overline{B}) = 65/100 = 0.65$. The frequency of $A \cap B$ is $20/100 = 0.20$; it is not a marginal frequency.

Example 2. A première class of 32 students is split as follows. Complete the margins and calculate the marginal frequencies "CS" and "girls".

	Boys	Girls	Total
CS	12	9	**21**
No CS	6	5	**11**
Total	**18**	**14**	32

Solution. $f(\text{CS}) = 21/32 = 65.625\%$; $f(\text{girls}) = 14/32 = 43.75\%$.

11.2 Conditional frequencies and reading sentences

Key point · Definition. The frequency of A **among the individuals of** B, if this subpopulation is nonempty, is

$$f_B(A) = \frac{\text{count satisfying } A \text{ and } B}{\text{count of } B}.$$

The denominator corresponds to the population introduced by "among".

Example 3. In the CS table, calculate the frequency of girls among the students without CS, then the frequency of students without CS among the girls. Are the two results equal?

Solution. Among the 11 students without CS, 5 are girls: $5/11 \approx 45.45\%$. Among the 14 girls, 5 do not take CS: $5/14 \approx 35.71\%$. The numerator is the same, the reference populations differ.

Example 4. In the table of 100 individuals, calculate $f_B(A)$, $f_A(B)$ and $f_{\overline{A}}(B)$. Translate each result into a sentence.

Solution. $f_B(A) = 20/35 = 4/7 \approx 57.14\%$ of the individuals of B satisfy A. $f_A(B) = 20/32 = 62.5\%$ of the individuals of A satisfy B. $f_{\overline{A}}(B) = 15/68 \approx 22.06\%$ of the individuals not satisfying A satisfy B.

Example 5. Does "60% of cyclists are students" allow us to conclude that 60% of students are cyclists?

Solution. No. For example, out of 10 cyclists, 6 are students; if the population has 100 students in all, only 6% of the students are cyclists.

11.3 Completing a table from frequencies

Example 6 · Fictional situation. A group has 200 people, of whom 120 use a bus (B). Among these 120 people, 75% have a pass (A). Among the other 80, 25% have a pass.

1. Construct the two-way table $B/\overline{B}$ and $A/\overline{A}$ with its margins.
2. Calculate the marginal frequency of A.
3. Calculate the frequency of B among A.

Solution. In B, $120 \times 0.75 = 90$ people satisfy A, so 30 do not. In $\overline{B}$, $80 \times 0.25 = 20$ satisfy A, and 60 do not.

	A	$\overline{A}$	Total
B	90	30	120
$\overline{B}$	20	60	80
Total	110	90	200

$f(A) = 110/200 = 55\%$. $f_A(B) = 90/110 = 9/11 \approx 81.82\%$.

Example 7. A headline announces "Three pass holders in four take the bus", based on $f_B(A) = 75\%$. Explain the error and propose a correct sentence.

Solution. The headline reverses the conditioning. The correct sentence is: "Three people in four among those who take the bus have a pass." The proportion of pass holders who take the bus is here about 81.82%.

11.4 Processing a file and collecting real data

```
# Exemple fictif : (transport, satisfaction)
donnees = [("bus", "oui"), ("marche", "oui"),
           ("bus", "non"), ("velo", "oui"),
           ("bus", "oui"), ("marche", "non")]

bus = [r for r in donnees if r[0] == "bus"]
bus_satisfaits = [r for r in donnees
                  if r[0] == "bus" and r[1] == "oui"]
autres = [r for r in donnees if not r[0] == "bus"]
```

Example 8. Give the sizes of the three lists. Calculate the frequency of satisfaction among bus users. Replace `and` by `or` in the second filter: how many rows are then selected?

Solution. Sizes: $3, 2, 3$. Among bus users, the frequency of satisfaction is $2/3$. With `or`, we select the 3 bus users and the 2 satisfied people using another means: 5 rows, without counting twice the rows satisfying both criteria.

Method · Activity with the class data. Collect anonymously two variables: main means of transport and journey time in ordered categories (< 15 min; from 15 to under 30 min; ≥ 30 min). One row per response, without names. Construct the two-way table, then two conditional frequencies in opposite directions. State the count, the collection date and any non-responses.

Example 9. Carry out the collection and write two justified conclusions. Explain why the results do not automatically describe all students in Madagascar.

Solution. The values depend on the actual collection. Check: one category per variable and per response; sum of the cells equal to the number of usable responses; correct denominator for each frequency; announced rounding. The sample is one particular class, not a representative sample of all students in the country. Non-responses must be distinguished from an observed category.